Question Paper ID: 571294

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Course Code: AI362IA

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)
R V Vidyanikethan Post
Mysuru Road, Bengaluru - 560059

VI Semester B.E. Regular / Supplementary Examination July / August 2026

Big Data Technologies

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Identify the characteristics of commodity hardware.	02	1	2
1.2	Identify the role of the secondary name node.	02	1	1
1.3	What is the role of YARN?	02	1	2
1.4	What is a Combiner Function?	02	2	2
1.5	Name any two SQL-on-Hadoop alternatives to Hive.	02	1	1
1.6	Differentiate between LOAD DATA LOCAL and LOAD DATA.	02	2	1
1.7	List the three core components of a Flume agent.	02	1	1
1.8	What is batching in Flume?	02	2	2
1.9	What is serialization in Spark?	02	1	2
1.10	What is the difference between Transformations and Actions?	02	2	2

Part B

Question No	Question	M	CO	BT
2a	Discuss the need for HDFS High Availability and summarize the consequences of NameNode failure.	08	1	2
2b	Illustrate the Anatomy of a File Read Operation.	08	2	3
3a	Discuss Map and Reduce phases in Hadoop MapReduce. dia	08	1	2
3b	Summarize Data Locality in Hadoop.	08	1	2

	OR			
4a	Discuss Hadoop Streaming and assess its effectiveness for implementing MapReduce programs compared to Java MapReduce.	08	2	4
4b	Analyze the role of Speculative Execution in improving the performance of Hadoop MapReduce jobs.	08	2	4
5a	Differentiate Embedded, Local and Remote meta store.	06	1	2
5b	A Wildlife Conservation Department conducts periodic surveys to monitor the population of different animal species across various regions. The department stores details such as AnimalID, Species, Region, Count, and Survey_Date in a Hadoop ecosystem. To analyze the wildlife census data efficiently, the department uses Apache Hive. Create a Hive table for the Wildlife Census dataset by considering suitable attributes. Write HiveQL queries for the following: i) Find the total animal count in each region. ii) Identify the most common species. iii) Display the regions where surveys were conducted during the last 3 years. iv) List the species found only in one region. v) Calculate the average animal count for each species.	10	3	3
	OR			
6a	Discuss the configuration and logging mechanisms in Hive.	08	2	2
6b	Compare SQL and HiveQL.	08	2	2
7a	Explain the configuration of a Spooling Directory Source and File Channel in Apache Flume with an example.	10	2	3
7b	Discuss, with a diagram and code snippet, the working of Flume.	06	3	2
	OR			
8a	A national banking system continuously collects ATM transaction logs from thousands of branches across the country using Apache Flume. To ensure uninterrupted data ingestion into Hadoop, the bank wants Flume to automatically switch to a backup HDFS sink whenever the primary sink becomes unavailable. Explain how Sink Groups can be configured in Apache Flume to achieve fault-tolerant data delivery. Illustrate your answer with a neat diagram and a suitable configuration.	10	3	2
8b	What are Multiplexing Selectors in Apache Flume? How do they route events to different sinks?	06	2	3
9a	Discuss Job submission phase of a Spark Job Run with a diagram?	10	2	2
9b	Explain Shared Variables in Apache Spark. Discuss the working of Broadcast Variables and Accumulators with suitable examples.	06	2	2
	OR			

10a	A smart agriculture system continuously records field temperatures to monitor crop conditions. Farmers require the maximum temperature of the day for irrigation planning. Write a Spark program in Scala to determine the maximum temperature from the recorded data and explain the execution flow.	08	3	2
10b	A university stores student marks in a text file. Write a Spark program in Scala to calculate the average marks of all students using RDD transformations and actions.	08	3	2